

1. Enumerate the various means of transport. Describe the considerations that lead to selection of a particular type of transport.

Answer:

The movement of people and goods from one place to another is known as transportation. There are various means of transport are as follows:

i. Highway:

- Highway is the most common means of transport and is most flexible and economical.
- It provides high accessibility to almost all potential destinations.
- Capital cost of physical facilities for highway is cheaper than other modes of transport.



ii. Railway:

- Railway is suitable for carrying large number of passengers and heavy goods over long distances.
- It provides moderate speeds and level of accessibility.
- Physical facilities for railway require heavy investment.



iii. Airway:

- Airway is the fastest mode of transportation mainly used for passengers and highly valuable goods.
- Accessibility in airway is limited but productivity is high due to high speeds.
- Capital and operating cost both are quite high for commercial air system.



iv. Waterway:

- Waterway includes transport through water bodies such as rivers, canals, lakes and sea using boats and ships.
- This mode of transport is economical for transporting heavy cargo.
- It is characterized by low speed and relatively low accessibility but extremely high capacities.

v. Ropeway:

- It is an important means of transportation in hilly and mountainous terrain.
- It utilizes cables to carry passengers and goods.
- Manakmana cable car, Hetauda-Kathmandu ropeway are some of the popular ropeways on Nepal.



vi. Pipeline:

- It is the means of transport used for transporting liquids and gases such as petroleum, natural gases or water through pipelines.
- It is characterized by low speed, high-capacity continuous flow transportation and involve large amount of working storage.

The selection of a particular mode of transport depends on the technical, economic, geographical and social factors.

The considerations that lead to selection of a particular type of transport are as follows:

i. Distance travelled:

- Roadway is suitable for short distance travel.
- Railway and airway are suitable for long distances travel or urgent travel.

ii. Nature of goods and passengers:

- Heavy and bulky goods are better transported by railway or waterway.
- Perishable goods require faster transport for which airway is suitable.

iii. Cost of transportation:

- The mode of transportation should be economical in terms of construction, operation and maintenance costs.
- Air transportation is most expensive means of transport while road transportation is economical than all the other modes.

iv. Speed:

- Urgent good transportation and travel favors air transport due to higher speed.
- Less urgent goods can be transported by railway or waterway which has lesser speed.

v. Topography:

- Ropeway, roadway or air transport may be suitable for the transportation facilities in mountainous region.
- Similarly, for the plain topographical areas, roadway and railways are suitable.

vi. Safety and reliability:

- Air transportation is considered the safest means of transport.
- The transport mode must ensure the safe movement of people and goods with minimum risk of accidents and delays.

2. **In the context of our difficult terrain and topography at one hand and economics situation at other, how could trail bridge network in our hilly region could play effective role to alleviate transportation access, elaborated with normal.**

Answer:

The complex rugged topography, deep river and ravines, scattered settlements in hilly and mountainous regions of Nepal makes the construction of road in every part of the country difficult and expensive. In such areas, trail bridges provide a practical and reliable means of transportation.

Trail bridges are basically the pedestrian suspension bridges that provide safe river crossings in rural and hilly areas. They are a major government priority under the Trail Bridge Sector Wide Approach Programme, guided by the Trail Bridge Strategy 2006 and the Trail Bridge Directives 2010.



Nepal has built over 9,500 trail bridges, total about 10,000 by 2023. Trail bridge sector is often described as a success story because it combines local participation, standardized technical manuals, and government-led implementation.

Trail bridge network in hilly region could play effective role to alleviate transportation access by:

- i. Improving connectivity that connects villages separated by rivers and gorges, access to schools, health centers and markets.
- ii. Enhancing access to social services such as education, health care etc.
- iii. Providing safe access to trekking routes and tourist destinations that promotes tourism and economic activities.
- iv. Ensures connectivity during floods and landslides, facilitating emergency rescues and relief operations.
- v. Reduces travel time and risk by eliminating dangers in river crossings.
- vi. Saves time and provides safety for the pedestrians and livestock.
- vii. Reduces regional disparities by linking remote communities with economic centers.
- viii. Causes less environmental disturbances compared to road construction.

3. **What are the steps that should be considered during the preparation of wearing course of a metaled highway?**

Answer:

Metaled highways in Nepal form the core of the country's transport network, constructed in major routes like the Mahendra Highway, Araniko Highway, and BP Highway. Wearing course is the topmost layer of the highway pavement, which provides smooth riding surface, resist traffic wear and tear and protects the underlying pavement from water and abrasion.



The steps that should be considered during preparation of wearing course of a metaled highway is listed in points below:

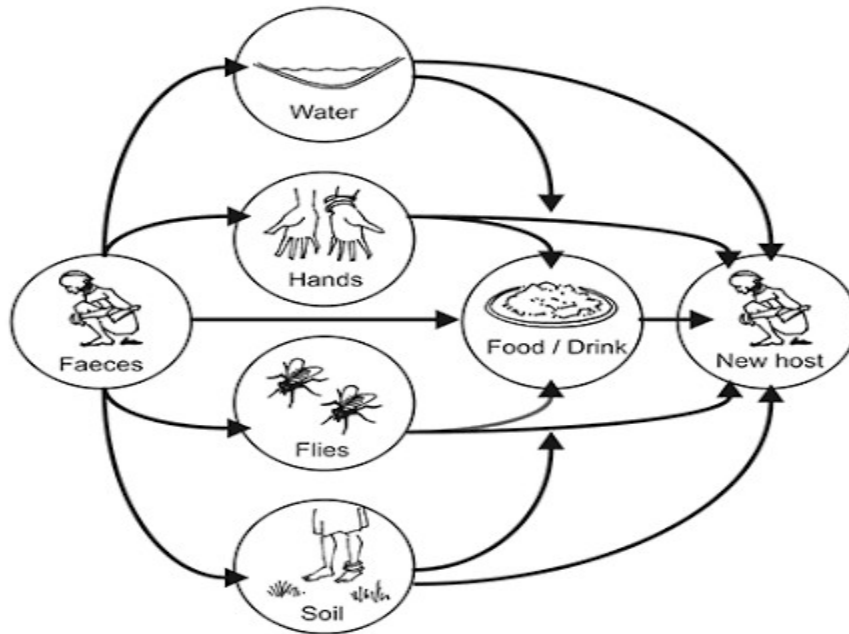
- i. **Inspection and Preparation of the Existing Surface**
 - a. At first, the condition of subgrade and base courses must be properly inspected.
 - b. The dust, loose materials, vegetation, and debris are thoroughly removed from the surface.
 - c. Even the potholes, cracks, and uneven areas on the surface is repaired.
- ii. **Surface Cleaning**
 - a. Then, the pavement is thoroughly cleaned using brooms, air compressors, or mechanical sweepers.
 - b. It must be ensured that the surface is dry and free from dirt before laying the wearing course.

- iii. **Application of Prime Coat or Tack Coat**
 - a. Now, a layer of prime coat is applied on granular bases or a tack coat on an existing bituminous surface.
 - b. This ensures proper bonding between the old surface and the new wearing course.
- iv. **Preparation of Bituminous Mix**
 - a. The bitumen and aggregates are then heated to the specified temperature.
 - b. They are uniformly mixed in a hot-mix plant according to the approved mix design.
- v. **Transportation of Mix**
 - a. The hot mix is then transported to the construction site in covered trucks to maintain the required temperature and prevent contamination.
- vi. **Laying the Wearing Course**
 - a. The mix is uniformly spread using a mechanical paver to the specified thickness, width, and cross slope (camber).
- vii. **Compaction**
 - a. The laid mix is compacted immediately using steel-wheel and pneumatic-tyred rollers.
 - b. The rolling is continued until the required density and smooth finish are achieved.
- viii. **Quality Control**
 - a. Now, the thickness, level, surface evenness, density, and temperature are checked.
 - b. It is to be ensured that the finished surface meets design specifications.
- ix. **Opening to Traffic**
 - a. The pavement is then allowed to cool and the finished surface is inspected.
 - b. The road is opened to traffic only after adequate compaction and cooling.

4. **Write down the different water borne diseases transmitted through polluted water. Explain their transmission routes and preventive measures.**

Answer:

- Waterborne diseases are those diseases caused due to consumption of water containing pathogenic microorganisms or chemical impurities.
- Water borne diseases are also known as water quality diseases.
- These diseases commonly occur using contaminated water during bathing, washing, drinking, preparation of food or consumption of food.
- The different water borne diseases transmitted through polluted water are listed below:
 - ✓ Typhoid (by salmonella typhi)
 - ✓ Cholera (by vibrio cholera bacteria)
 - ✓ Dysentery (by shigella bacteria)
 - ✓ Paratyphoid (by salmonella paratyphi)
- Faecal Oral Transmission
 - Organisms from the excreta of animals reaches to water sources directly or sewage disposal.
 - Then this water is transmitted in the human body orally from the food or contaminated water
 - This process is known as faecal oral transmission.
 - Water washed and water borne diseases is caused by faecal oral route.
 - Transmission of the water-related disease is related to water supply and sanitation due to inadequate and improper disposal of waste and unsafe drinking water.
 - The diagram of faecal oral transmission is shown below:



➤ Preventive measures:

- i. Proper washing of hands after the use of toilet and before meal.
- ii. Proper covering of food and water during storage.
- iii. Water must be fetched from clean water sources.
- iv. Sewage should not be disposed at or near the water resources.
- v. Only properly filtered water must be drunk.
- vi. Avoid fly nuisance by proper disposal of animals and organic wastes.
- vii. Public awareness and improving the living habits of people.

5. **Draw a flow chart of each component of a typical rural water supply system and explain the function of each component.**

Answer:

In a gravity water supply system in rural hilly area of Nepal, source is usually fast flowing rivers and springs which is relatively free from harmful suspended materials. Hence, treatment plant is not incorporated in water supply project in rural hilly areas. However, in a gravity water supply system in rural terai region, treatment plant must be incorporated to remove the suspended and other harmful materials from the water drawn from the source.

A flowchart of typical gravity Water Supply System (RWSS) in rural area of Nepal is shown below:

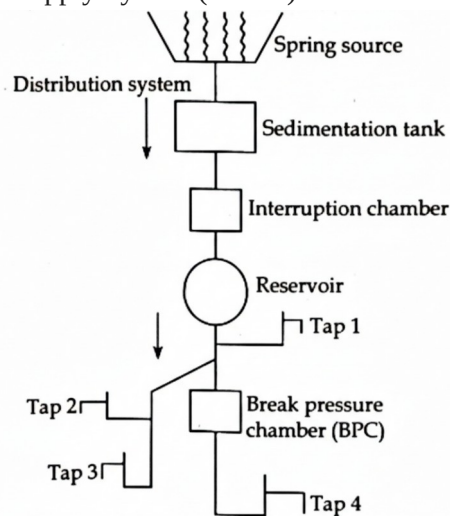


Figure: Typical rural water supply system

Components of water supply systems:

- i. Source of water:
 - Source of water should be reliable and contain minimum impurities.

- The amount of water available in the source must be adequate to tap the required amount to supply it to consumers.
 - For rural hilly areas, source of water is springs which is considered to be relatively pure hence no treatment plant is provided in the water supply project of such areas.
- ii. Intake:
 - It is a device or structure constructed at or near the water source to draw the required amount of water to supply line.
 - Must be easily accessible and must have minimum possibility of contamination.
 - Primary function of intake is to draw the required amount of water from source.
 - iii. Distribution chamber:
 - The main function of distribution chamber is to facilitate the divisions of flow in distribution subsystems.
 - iv. Transmission mains:
 - The main function of transmission mains is to convey the water from source to treatment plant.
 - v. Interruption chamber:
 - Interruption chamber is provided in transmission mains to break the pressure without float valve to prevent bursting of pipe due to excess pressure.
 - vi. Reservoir:
 - The main function of reservoir is to store water from the source during low demand period and utilize it during high demand.
 - vii. Valve chamber:
 - Chamber constructed to protect the valve in water supply projects.
 - viii. Distribution system:
 - It is the pipe network to deliver water to consumer from the distribution reservoir.
 - ix. Break pressure tank:
 - It is a pressure releasing device to break the pressure with float valve to prevent bursting of pipe due to excessive pressure.
 - Provided in distribution mains.

6. **In a view of emerging crisis in energy sector, what are your suggestions in the context of alternate energy system which ought to be developed in Nepal so that we do not depend solely on conventional system?**

Answer:

The over-exploitation of non-renewable energy sources such as coal, petrol, diesel etc. has resulted in emerging crisis in energy sector globally. The price hikes and fuel shortages has hampered in the overall global economy. Not only the economic sector, energy crisis has impacted the day to day living of people and environmental aspects too. Especially in the mountainous country like Nepal, which is among the high risk of climate change, we ought to develop the alternate energy system rather depending solely on the conventional system. The culture of preferring alternate energy is gradually rising in Nepal. Nepal is experiencing one of the fastest electric vehicle adoption rates globally, with EVs comprising roughly 73% of private passenger four-wheeler imports. Not only that, people of Nepal opting for induction chulos, biogas rather than butane gas cylinders also shows the increased attraction towards alternate energy systems.

Alternate energy system refers to the energy obtained from renewable and environmentally friendly sources. In Nepal, alternative energy plays a vital role in providing clean, reliable, and affordable energy, especially in rural and remote areas. As Nepal is a rich country in terms of natural resources, there is a humungous scope of alternate energy generation. There is an independent government institution called Alternative Energy Promotion Centre (AEPIC) under Ministry of Energy, Water Resources and Irrigation. AEPIC supports technologies such as solar, biogas, small hydro, improved cookstoves, and other decentralized renewable energy systems. Its goal is to improve energy access, especially in rural areas, while reducing environmental impact.

My suggestion in the context of promotion of alternate energy systems in Nepal would be the optimum utilization of natural resources for economical and sustainable use of alternate energy. The major alternative energy systems that can be suitable and economical to develop in Nepal are listed below:

i. Hydropower:

- It is the electricity generated by utilizing the energy of flowing rivers.
- Globally, hydropower covers about 14.3% of total electricity generation.
- Nepal has more than 6000 big and small rivers which has net electricity generation capacity of 83000 MW making it a second richest country in water resources after Brazil.
- Nepal currently has reached 4200 MW of installed electricity capacity, of which over 90% is hydropower.
- Due to steep topography that provides the required head for power generation and availability of fast flowing rivers in Nepal, hydropower is most suitable option of alternate energy system.



ii. Solar energy:

- It is the electricity generated by utilizing the sunlight.
- Powered by the photovoltaic effect (using silicon solar cells) or solar-thermal technology, it provides a renewable, clean, and increasingly cost-effective way to power homes, businesses, and entire utility grids.
- Solar energy covers about 10% of global electricity demand.
- It is ideal for remote mountain regions with high solar radiation where other alternate source of energy is difficult to supply or are expensive.



iii. Biogas:

- Biogas is a combustible gas produced by the breakdown of organic materials such as agricultural residues, animal manure, food waste, and sewage by microorganisms in an oxygen-free environment
- Biogas covers a very small fraction of total national energy consumption, accounting for less than 1%.
- It is very popular in rural households where agricultural wastages such as cow dungs can be utilized to produce cooking gas.
- Nepal has successfully generated and utilized locally produced green hydrogen to power hydrogen cookstoves of which the pilot facility was officially inaugurated in Municipality, Baglung, on 15th May 2026.



iv. Wind energy:

- Wind energy is a renewable power source that utilizes the kinetic energy of moving air to generate electricity.
- Potential of wind energy exists in the district of Nepal with the presence of high velocity wind such as Mustang, Palpa, Dolpa, and other high-wind regions.
- It is currently limited in application.



7. What are the reasons that farmers managed irrigation system in Nepal seems to be pragmatic as well as cost effective? How the coverage of irrigation could be widened in Nepal so that farmers could be benefitted?

Answer:

- **Farmer Managed Irrigation Systems (FMIS)** are irrigation schemes that are developed, operated, and maintained entirely by the farming communities they serve. In the context of Nepal, these systems represent a significant part of the agricultural landscape and have a long historical tradition.
- Before 1922, irrigation in Nepal was primarily developed and managed through FMIS. It was only after this period that the government began making significant efforts to develop state-led irrigation infrastructure.
- FMIS are the backbone of Nepal's irrigated agriculture, covering approximately **70% of the country's total irrigated land**.
- Some of the biggest FMIS maintained in Nepal are Rajapur Irrigation System, Rani Jamara Kulariya Irrigation System etc.



- The reasons that farmers managed irrigation system in Nepal seems to be pragmatic as well as cost effective are listed in points below:
 - i. Strong community participation and ownership.
 - ii. Low operation and maintenance cost due to farmers' contributions.
 - iii. Efficient and equitable water distribution based on local rules.
 - iv. Quick repair and maintenance because users are directly involved.
 - v. Uses local knowledge and indigenous technology.
 - vi. Improves agricultural productivity and rural livelihoods.
 - vii. Promotes social cooperation and conflict resolution among farmers.

viii. Sustainable due to users' sense of responsibility.

The irrigation could be widened in Nepal to benefit the farmers by:

- i. Developing new irrigation projects, prioritizing the irrigation development in water-scarce areas.
- ii. Repairing and rehabilitation of existing irrigation systems, old canals, headworks etc.
- iii. Promoting groundwater irrigation by installing tube wells in Terai region
- iv. Promoting drip and sprinkler irrigation to save water and increase irrigation efficiency.
- v. Strengthening Farmers Managed Irrigation System.
- vi. Improving water management by canal lining or introduction of rotational water distribution.

8. **Explain the objective of river training works. Explain briefly the different methods of river training works.**

Answer:

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries.

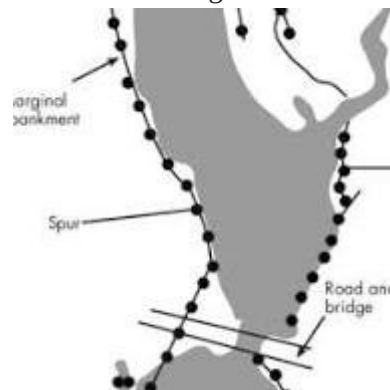
The objectives of river training works are as follows:

- i. To protect river banks from erosion by preventing scouring and collapse of river banks.
- ii. To protect hydraulic structures and ensures safety of bridges, dams, weirs and other structures from erosion and scour.
- iii. To provide safe waterway for navigation by maintaining sufficient depth and stable channel.
- iv. To reduce flood damage by regulating the river flow.
- v. To minimize silt accumulation near bridges, intakes etc.

There are different methods of river training works. Some of them are described below:

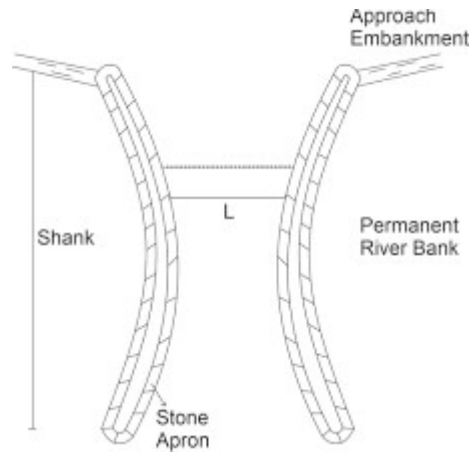
i. **Marginal embankments/ levees:**

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



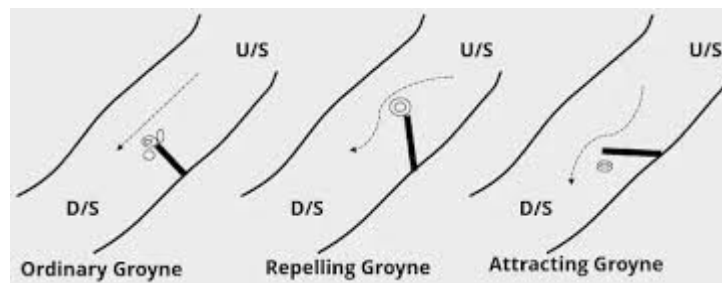
ii. **Guide banks:**

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



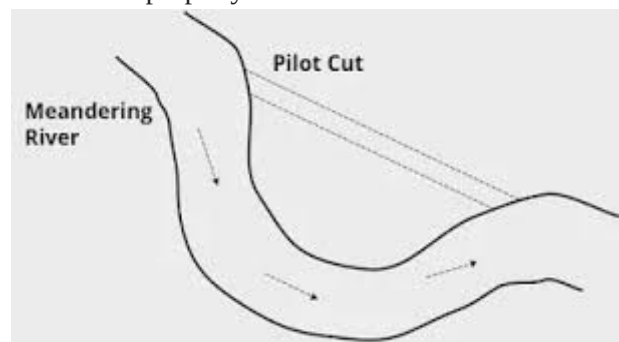
iii. Groynes or spurs:

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



iv. Artificial cutoff:

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.



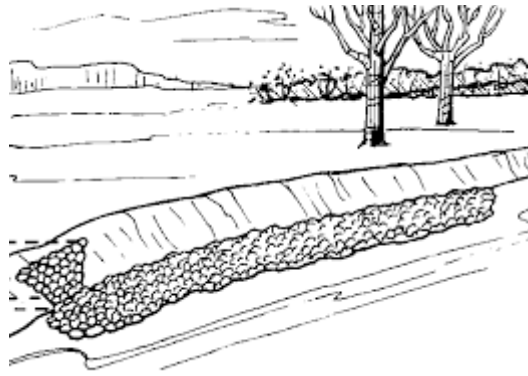
v. Bank protection and Pitched banks:

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

Bank protection work can be classified as direct or indirect.

Direct protection works include work done on the bank itself such as vegetation cover, pitching

Indirect protection includes work constructed not directly on the banks but in front of them.



vi. Pitched islands:

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



vii. Bandalling:

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.



9. What are the roles of National Building codes in Nepal? How does the code address the problem of earthquake? How could the code be made effective?

Answer:

- National Building Code (NBC) of Nepal provides minimum standards for the planning, design, construction, and maintenance of buildings to ensure safety, durability, and quality.
- Nepal's National Building Code was drafted in the 1990s, approved by the government in 2003, and legally enforced from 2005 in municipalities.
- A more recent update, NBC 205: 2024, has also been issued by the Department of Urban Development and Building Construction.
- The roles of national building codes in Nepal are listed below:
 - i. NBC ensures structural safety of the buildings constructed in Nepal.
 - ii. It reduces the risk of damage from earthquakes, fire, wind, and other hazards.
 - iii. It promotes the uniform standards for building design and construction.

- iv. The code guides engineers, architects and builders in safe construction practices.
- v. NBC also improves the quality and durability of buildings.
- vi. It regulates the use of appropriate construction materials and methods.
- vii. NBC also supports the sustainable and planned urban development.

Nepal lies in a highly seismic zone; Hence, NBC incorporates earthquake-resistant design provisions. In Nepal's National Building Code, earthquake-resistant design is mainly covered under NBC 105, which sets the rules for seismic analysis and design of buildings.

The code addresses the problem of earthquake by:

- i. Specifying the earthquake-resistant structural design for RCC, steel, masonry and timber buildings.
- ii. Requiring the ductile detailing of reinforced concrete members to improve energy absorption during earthquakes.
- iii. Providing minimum standards for foundations, materials and construction quality.
- iv. Requiring proper reinforcement detailing at beam-column joints.
- v. Recommending proper site selection by avoiding unstable slopes, liquefiable soils and fault zones whenever possible.

The code could be made effective by:

- i. Strict enforcement of code by local governments.
- ii. Making building permits conditional on compliance with the prevailing code.
- iii. Training engineers, architects, contractors in NBC provisions.
- iv. Updating the code regularly based on new research and lessons from earthquakes.
- v. Promoting seismic assessment and retrofitting of existing vulnerable buildings.

10. Name the various steps involved in the “Environmental impact assessment “of a project and explain the function of each step.

Answer:

Environmental Impact assessment (EIA):

- "Environmental impact assessment" is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.
- Projects requiring EIA are listed in schedule 3 of EPR 2077.
- Includes:
 - Screening → Scoping → Preparation and approval of work schedule → Field study and public hearing → EIA Report Preparation → Recommendation and submission → final review and approval → Decision making
- EIA is required for large-scale development projects with significant environmental impacts that cannot be adequately assessed through a Brief Environmental Study (BES) or Initial Environmental Examination (IEE).
- EIA is done for:
 - a. Highways with the length of road greater than 50 km
 - b. Irrigation project with command area greater than 2000 hectares.
 - c. Hydropower projects with installed capacity of greater than 50 MW.

Stepwise procedure for EIA study and its approval:

i. Screening:

- This step determines whether the proposed project requires IEE or EIA.
- No project requiring EIA can be implemented without approval of the environmental study report.

ii. Preparation of Scoping Document:

The proponent prepares Scoping Document following the format in **Schedule-5** and is submitted to the concerned approving authority.

- For federal projects, it is submitted to the **Concerned Ministry** (e.g., the Ministry of Irrigation or Physical Infrastructure).
- The Concerned Ministry reviews it, adds its opinion, and forwards it to the **Ministry of Forests and Environment (MoFE)** within **15 days**.
- MoFE must approve or modify the Scoping Report within **15 days** of receiving the recommendation.

iii. Preparation and Approval of work schedule:

- The proponent prepares a **Work Schedule** (per **Schedule-8**) that outlines the methodology and timeline for the study.
- The Scoping Report and Work Schedule are often submitted and approved together.
- Once reviewed, MoFE must approve the Work Schedule within **15 days**.

iv. Field study and Public Hearing:

- The study must be carried out by a team of qualified experts who meet the academic and experience criteria listed in **Schedule-13**.
- The proponent **must organize at least one public hearing** in the project area to gather feedback from local communities, forest user groups, and local representatives.
- The proponent **must publicize the hearing** via local radio, newspapers, or other media, and post notices at the ward office. Documentation of the hearing (attendance, suggestions, photos, and audio-visual recordings) must be included in the final report.

v. EIA Report Preparation:

- EIA report is prepared based on the approved Scoping Report, Work Schedule, and the suggestions from the public hearing according to the detailed format in **Schedule-12**.
- **Public Notice for Feedback:** The proponent must issue another **7-day notice** (Schedule-9) in a national daily newspaper to gather written suggestions on the potential environmental impacts of the project.
- Reports are generally prepared in **Nepali** language.
- For foreign-invested projects, they may be in English, provided a concise **executive summary in Nepali** is included.

vi. Recommendation and Submission:

- The proponent must include a formal **Recommendation Letter (सिफारिस पत्र)** from the concerned **Local Level** and the relevant thematic office (per **Schedule-14**). These bodies must provide the recommendation within **15 days**.
- **Submission:** For federal projects, the report is submitted to the **Concerned Ministry**.
- The Concerned Ministry reviews the report and forwards it to **MoFE** with its formal opinion within **15 days**.
- The final report must be submitted within **two years** of the Scoping/Work Schedule approval.

vii. Final Review and Approval of EIA:

- **Final Public Review:** Upon receiving the EIA report, MoFE must make it public for **7 days** on its website and via a national daily newspaper to collect final public comments.
- **Review Committee:** MoFE may form a **committee** of representatives and experts to investigate the report, which may include on-site inspections or monitoring.
- **Approval Decision:** If the investigation shows the project will not cause significant adverse environmental impacts, MoFE shall approve the EIA report within **35 days**. The approval may include specific, binding conditions that the proponent must follow.
- Additionally, the project must start within **three years** of approval, or the proponent may be required to conduct a new study.

Note on Violations: Implementing a project without an approved EIA report or in violation of its conditions can lead to a fine of up to **5 million (50 लाख) rupees**.